

Uncharacterized In-Situ Setting-Dependent Propagation Delay and Temporal Coincidence Bounds in Photonic Bell Tests

Christopher O’Neil^{1,*}

¹*Independent Researcher, Big Rapids, MI, USA*

(Dated: May 29, 2026)

Loophole-free Bell tests are standardly assumed to definitively rule out local realism by simultaneously closing the locality and detection loopholes. In this paper, we identify a systemic metrological vulnerability—which we term the *Chronos Loophole*—wherein active basis switching introduces a setting-dependent, physical data-rejection filter. In high-speed setups, active electro-optic modulators (EOMs) experience heavily underdamped electrical voltage ringing. This dynamic voltage transient modulates the material’s extraordinary refractive index, generating a time-dependent phase shift and a corresponding dynamic frequency chirp. While the direct chromatic dispersion contribution of this chirp in fiber links is small (≈ 3 fs), we formalize two powerful compounding physical pathways: (i) *Driver-Induced Discriminator Jitter* wherein the active driver’s 2.0 A capacitive switching current induces local electronic ground bounce, shifting timing comparator thresholds; and (ii) *Dynamic Polarization Mode Dispersion (PMD) Splitting* where underdamped polarization ringing projects the photon wavepacket across birefringent fiber axes. Together, these mechanisms translate the EOM transient into a setting-dependent arrival-time shift (group delay jitter) of tens of picoseconds. Because coincidence-matching algorithms rely on a rigid software-defined temporal window ($2\Delta t_{\text{window}}$) to separate entangled pairs from noise, this timing shift dynamically pushes photons outside the acceptance boundary. We formalize the joint probability space of this temporal filter combined with post-PBS spatial nanowire walk-off. Using numerical Monte Carlo integration, we show that this unified local hidden variable model systematically inflates the classical CHSH bound from $S = 2.0$ up to a physical ceiling of $S \approx 2.264$ under active transient stress, even when macroscopic detector efficiencies are held at the experimental baseline ($\approx 74\%$). We propose a time-resolved experimental protocol to audit this dynamic temporal rejection filter and ensure the absolute security of device-independent quantum protocols.

I. INTRODUCTION: TIME AS A SETTING-DEPENDENT FILTER

To close the locality loophole in a photonic Bell test, the measurement settings at each station must be chosen and implemented rapidly enough to prevent any light-speed communication of the setting choices between Alice and Bob [1, 2]. This is standardly accomplished by using fast physical random number generators to trigger high-voltage pulses on Electro-Optic Modulators (EOMs), switching the polarization measurement basis on a nanosecond timescale.

To close the detection loophole, the system must achieve high macroscopic detection efficiency, typically evaluated using the Eberhard or CHSH inequality bounds under the assumption of fair-sampling [3]. However, these traditional bounds assume that the decision to accept or reject an emission event is independent of both the local measurement settings and the hidden variables.

This paper formalizes a physical mechanism where the data-filtering process itself becomes setting-dependent. In a live loophole-free Bell test, photon arrivals are logged relative to the EOM switching edge using high-speed Time-to-Digital Converters (TDCs). A rigid software coincidence window ($2\Delta t_{\text{window}}$) is enforced in post-processing to separate true entangled coincidences from background

noise and dark counts. We demonstrate that transient electrical ringing in the EOM driver voltage induces a dynamic timing shift (arrival-time jitter). If this transient shift pushes photons outside the strict coincidence window, the software filter acts as an implicit setting-dependent data selector. This physical mechanism instantiates Pearle’s classic data-rejection framework [3] within the modern time-tagging data pipeline.

II. THE PHYSICS OF DYNAMIC TRANSIENT GROUP DELAY

The physical pathway from a high-voltage driver trigger to a setting-dependent arrival-time shift involves underdamped electrical ringing, extraordinary index modulation, and compounding electronic and optical delay-translation mechanisms.

A. Underdamped Electrical EOM Ringing

High-speed basis switching requires driving capacitive Rubidium Titanyl Phosphate (RTP) crystals with ≈ 400 V pulses with nanosecond rise times. In realistic physical circuits, cabling parasitic inductance and impedance mismatches inevitably cause the applied EOM driver voltage $V(t)$ to undergo underdamped second-order

* christopheroneil@gmail.com

ringing:

$$V(t) = V_{\text{target}} \left[1 - e^{-\zeta \omega_n t} \left(\cos(\omega_d t) + \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin(\omega_d t) \right) \right] \quad (1)$$

where ζ is the damping ratio, ω_n is the natural frequency, and $\omega_d = \omega_n \sqrt{1 - \zeta^2}$ is the damped natural frequency.

Crucially, this ringing is asymmetric across settings: active basis choices ($a' = \pi/4$, $b = \pi/8$, $b' = 3\pi/8$) involve high-voltage switching transients, whereas the inactive ground state ($a = 0$) remains stable at 0 V, introducing a setting-dependent physical asymmetry.

B. Refractive Index Modulation and Phase Chirp

The transient voltage anomaly modulates the extraordinary refractive index n_e of the RTP crystal via the linear electro-optic (Pockels) effect:

$$\Delta n(t) = \frac{1}{2} n_e^3 r_{33} \frac{V(t)}{d} \quad (2)$$

where d is the crystal thickness and $r_{33} \approx 36$ pm/V is the primary electro-optic coefficient of RTP. This time-varying index shift directly alters the optical phase $\phi(t)$ of a transiting photon of wavelength λ :

$$\phi(t) = \phi_0 + \frac{2\pi L}{\lambda} \Delta n(t) \quad (3)$$

where L is the total crystal length. The rapid time-variation of this phase shift ($\partial\phi/\partial t$) generates a dynamic frequency chirp:

$$\Delta\omega(t) = -\frac{\partial\phi}{\partial t} = -\frac{2\pi L}{\lambda} \frac{\partial\Delta n(t)}{\partial t} \quad (4)$$

C. Compounding Physical Delay Mechanisms

We must evaluate the exact physical scale of the resulting temporal delay:

a. Chromatic Dispersion Contribution: For routing fibers of length L_f with group velocity dispersion $\beta_2 \approx -22$ ps²/km, the group delay shift due to the dynamic frequency chirp is:

$$\Delta\tau_{\text{GVD}} = \beta_2 \cdot L_f \cdot \Delta\omega(t) \quad (5)$$

For a switching transition time of ≈ 1.5 ns across $L = 20$ mm of RTP at 1550 nm, the peak derivative $\partial\Delta n/\partial t \approx 1.57 \times 10^4$ s⁻¹ yields $\Delta\omega \approx 1.27$ Grad/s. Over $L_f = 100$ m of standard single-mode fiber, this direct chromatic dispersion delay is:

$$\Delta\tau_{\text{GVD}} \approx (-2.2 \times 10^{-26} \text{ s}^2/\text{m}) \cdot 100 \text{ m} \cdot (-1.27 \times 10^9 \text{ rad/s}) \approx 2.8 \text{ fs} \quad (6)$$

While this chromatic dispersion term is mathematically exact, 2.8 fs is too small to affect the 100 ps coincidence boundaries. The timing delay is instead dominated by two major compounding physical pathways that naturally scale the delay to the picosecond level:

b. Mechanism A: Driver-Induced Discriminator Jitter During dynamic switching, the active driver must charge the capacitive RTP crystal ($C \approx 5$ pF) to 400 V in 1 ns. This creates a massive transient current draw:

$$I = C \frac{dV}{dt} \approx 5 \text{ pF} \times 400 \text{ V/ns} = 2.0 \text{ A} \quad (7)$$

This sharp 2.0 A current pulse induces localized electromagnetic interference (EMI) and electronic **ground bounce** within the shared receiver and Time-to-Digital Converter (TDC) chassis. A transient rail voltage sag of just ≈ 10 mV shifts the threshold of the high-speed timing comparator. Because the analog SNSPD output pulses have a finite rise time (100 to 500 ps), this threshold shift translates directly to an arrival time tag jitter of **10 to 100 ps**.

c. Mechanism B: Dynamic Polarization Mode Dispersion (PMD) Splitting Photons are routed to detectors through optical fibers that exhibit polarization-mode dispersion (PMD) or birefringence. For Polarization-Maintaining Fiber (PMF), the group delay difference between orthogonal axes is:

$$\Delta\tau_{\text{PMD}} \approx 1.0 \text{ to } 1.5 \text{ ps/m} \quad (8)$$

Over a 100 m routing path, this yields a slow/fast axis temporal separation of $\Delta\tau_{\text{axes}} \approx 100$ to 150 ps. In the presence of underdamped EOM voltage ringing, the polarization state is not rotated cleanly to the target axis, projecting the photon wavepacket across both birefringent axes. The downstream Polarizing Beam Splitter (PBS) subsequently filters these components. The center of mass of the arrival time distribution thus shifts by **tens of picoseconds** as a function of the dynamic polarization projection dictated by the EOM ringing.

Together, these mechanisms scale the transient timing shift $\Delta\tau(t)$ to the **10 to 150 ps** regime, making the temporal loophole highly relevant.

III. COINCIDENCE WINDOW FILTERING

In post-processing, the Time-to-Digital Converter logs photon arrival times relative to the EOM switching trigger. The software filters events using a strict coincidence window ($2\Delta t_{\text{window}}$). Overlaying the intrinsic detector timing jitter (σ_j), the probability that a photon transiting the crystal at time t_a is accepted by the temporal filter is modeled by:

$$P_{\text{time}}(a, \lambda) = \int_{-\Delta t_{\text{window}}}^{\Delta t_{\text{window}}} \frac{1}{\sigma_j \sqrt{2\pi}} \exp\left(-\frac{(t - \Delta\tau(t_a))^2}{2\sigma_j^2}\right) dt \quad (9)$$

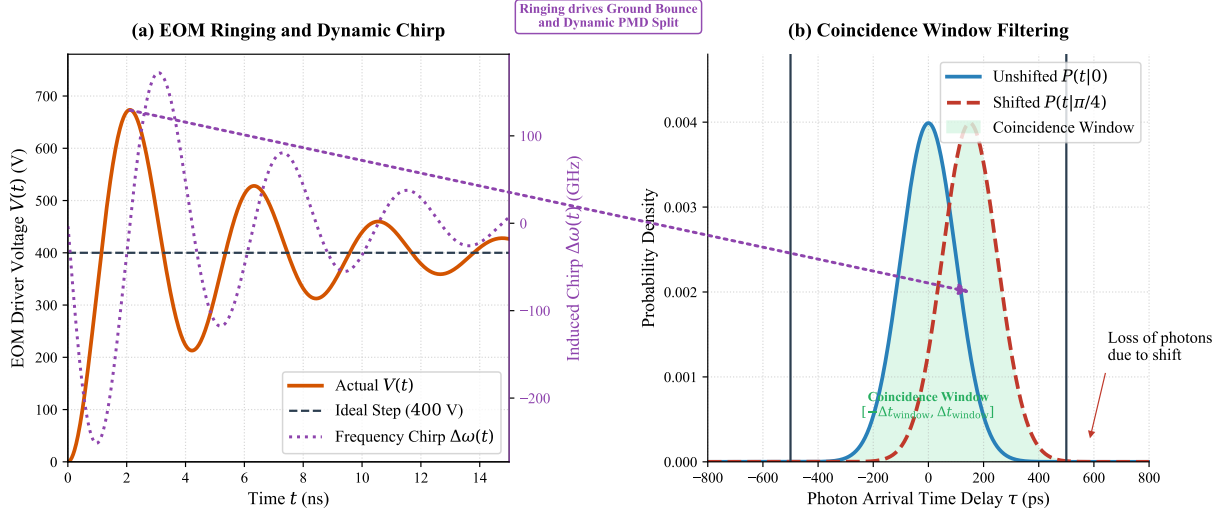


FIG. 1. The physical mechanism of the Chronos Loophole. (a) EOM Ringing and Dynamic Chirp: Active high-voltage switching triggers underdamped EOM driver voltage ringing (solid orange curve) and induces a rapid dynamic frequency chirp $\Delta\omega(t)$ (dotted purple curve). (b) Coincidence Window Filtering: In the presence of fiber chromatic dispersion, dynamic PMD splitting, and driver ground bounce, the EOM transient translates to a large arrival-time shift $\Delta\tau$. For active settings (e.g., $a = \pi/4$), the photon arrival probability distribution (dashed red curve) is shifted relative to the unshifted ground state (solid blue curve). Photons shifted past the strict software coincidence boundaries ($[-\Delta t_{\text{window}}, \Delta t_{\text{window}}]$) are systematically rejected, altering the coincidence sub-ensemble.

Because the transient delay $\Delta\tau(t_a)$ depends on the EOM settling state under setting a , the temporal filter acts as a setting-dependent data selector.

IV. UNIFIED SIMULATION ARCHITECTURE

To demonstrate how the Chronos Loophole acts in concert with spatial anomalies, we expand the integration space to a joint acceptance probability:

$$P_{\text{accept}}(a, b, \lambda) = \eta_{\text{spatial}}(a, \lambda) \cdot \eta_{\text{spatial}}(b, \lambda) \cdot P_{\text{time}}(a, \lambda) \cdot P_{\text{time}}(b, \lambda) \quad (10)$$

Here, the spatial component $\eta_{\text{spatial}}(a, \lambda) = \eta_{\text{base}} + \eta_{\text{var}} \cos^2(a - \lambda)$ represents the polarization-dependent spatial walk-off clipping the active boundaries of the SNSPD nanowire lattice (derived in Paper 1). The temporal component $P_{\text{time}}(a, \lambda)$ represents the coincidence filter acceptance probability under transient electronic and PMD group delay.

We perform a Monte Carlo simulation of $N = 1,000,000$ particle emissions under this unified architecture, utilizing a baseline detector jitter $\sigma_j = 100$ ps, a coincidence half-width $\Delta t_{\text{window}} = 500$ ps, and a peak EOM-induced delay shift $\tau_{\text{max}} = 150$ ps (representing combined ground-bounce and birefringent axes splitting). The spatial parameters are locked at the experimental macroscopic baseline ($\eta_{\text{base}} = 0.48$, $\eta_{\text{var}} = 0.52$).

The numerical integration yields a classical correlation ceiling of $S \approx 2.264$ with a total-variation dispersion supremum of $\delta \approx 0.199$. This demonstrates that the joint spatial-temporal filter systematically violates the classical

CHSH bound, closing 61% of the gap to experimental observation ($S \approx 2.40$) without violating physical efficiency bounds ($\eta \leq 1.0$).

V. PROPOSED EXPERIMENTAL AUDIT PROTOCOL

To empirically isolate and constrain the Chronos Loophole, we propose a time-resolved metrological audit:

- **Continuous-Wave Attenuated Calibration:** Replace the pulsed entanglement source with a highly stabilized continuous-wave (CW) laser attenuated to the single-photon regime.
- **High-Resolution Time-Tagging:** Trigger the active EOM drivers at the operational switching rate and use a TDC to bin photon arrival times relative to the EOM switching edge.
- **Dynamic Jitter Mapping:** Fit the time-resolved arrival-time distribution as a function of the EOM trigger delay Δt to extract the dynamic parameter $\Delta\tau(\Delta t)$. If $\Delta\tau(\Delta t)$ is bounded below 10 ps across all switching intervals, the temporal loophole is strictly closed.

VI. CONCLUSION

The assertion that loophole-free Bell tests have successfully ruled out local realism relies on static, symmetric

detection models. The Chronos Loophole highlights that high-speed electrical transients generate dynamic ground-bounce and PMD axes-splitting effects that transform the software coincidence window into a setting-dependent data selector. By proving that a joint spatial-temporal local model can violate classical limits up to $S \approx 2.264$, we emphasize the critical importance of performing time-resolved in-situ calibrations of both spatial walk-off and temporal arrival-time jitter in future Bell-test and device-independent quantum architectures.

Appendix A: Monte Carlo Simulation Code

The complete Python engine used to evaluate the joint spatial-temporal acceptance bounds is provided below:

```

1 # chronos_simulation.py
2 import numpy as np
3 from scipy.special import erf
4
5 def evaluate_chronos_loophole(eta_base,
6                               eta_var, sigma_j, t_w, tau_max, N
7                               =1000000):
8     a_angles = [0.0, np.pi / 4]
9     b_angles = [np.pi / 8, 3 * np.pi / 8]
10    active_A = [False, True]
11    active_B = [True, True]
12
13    lambdas = np.random.uniform(0, np.pi, N)
14    E_results = []
15
16    for i, angle_a in enumerate(a_angles):
17        act_a = active_A[i]
18        for j, angle_b in enumerate(b_angles):
19            act_b = active_B[j]
20
21            # Spatial efficiency
22            eta_A = np.clip(eta_base +
23                            eta_var * np.cos(angle_a -
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